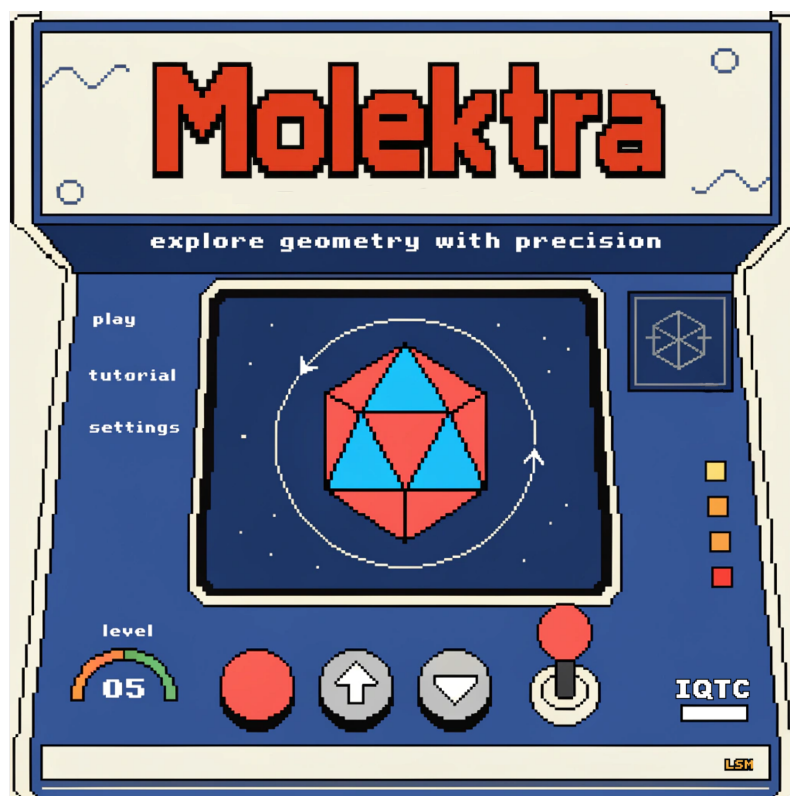


Molektra



Molecular visualization tool

User manual version 1.0

Laura Sánchez-Muñoz, Jordi Cirera

Table of Contents

1. Introduction	3
2. Software	3
3. <i>Molektra</i> Main Interface	4
4. Determination of the Continuous Shape Measures (CShM)	8
5. Point Group and Continuous Symmetry Measures (CSM)	11
6. Keyboard Shortcuts Summary	13
7. List of reference polyhedra	13
8. How to cite	16
9. References	16

1. Introduction

The Continuous Shape Measures (CShM) and Continuous Symmetry Measures (CSM) are powerful methodologies that have been widely used in computational chemistry and coordination chemistry over the last 20 years to quantify deviations from ideal molecular geometries and symmetries. These approaches provide with a rigorous numerical framework for comparing real molecular structures against ideal polyhedral or symmetry reference models, allowing researchers to analyze structural distortions in a precise and systematic way. Despite their usefulness, the practical application of these methods has traditionally relied on command-line programs, which often require manual preparation of input files, specialized syntax and computer skills, and interpretation of non-visual numerical outputs. Such limitations can make the analysis process difficult for user, less intuitive, and less accessible to the broader audience.

To address these challenges, the present software has been developed as an interactive environment that integrates molecular visualization with symmetry analysis and continuous shape measures. By integrating CShM and CSM methodologies within a visual and user-friendly platform, this software allows for the visualization, atom selection, geometric measurements, symmetry determination, and result interpretation of a given molecular structure. The software combines real-time 3D molecular rendering with automated analysis tools, allowing the user to perform advanced structural studies more efficiently while reducing the complexity associated with traditional console-based approaches.

2. Software

2.1. Software Setup

To ensure that the graphical interface and all its scientific dependencies work correctly, the software must be downloaded in the version corresponding to your operating system (Windows or macOS). If you are using macOS, please note that a different version is required depending on whether your device is equipped with an M1/M2 (Apple Silicon) or Intel chip. Additionally, macOS security defaults do not usually allow you to open downloaded software. To bypass macOS security blocks on

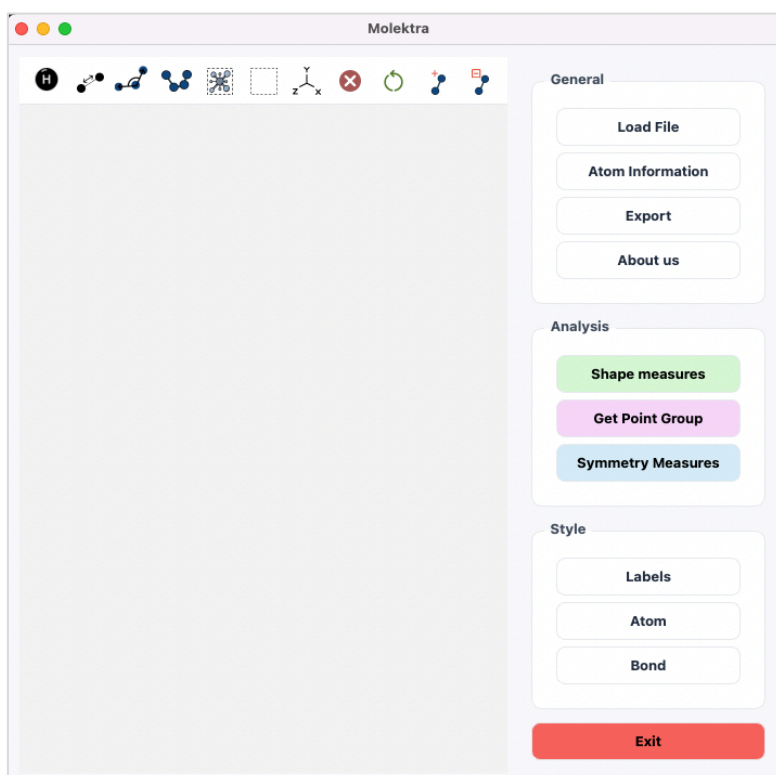
downloaded files, right-click (or Control-click) the file, select Open, and click Open Anyway.

2.2. Calling Molektra

Double-click the executable file.

3. Molektra Main Interface

If installation has been successful, you should encounter the Molektra main window, which contains the following options



3.1 Main Window Overview

When the program starts, the main graphical interface is displayed. This window provides access to all visualization, measurement, and symmetry-analysis tools available in the software.

The interface is divided into two main sections:

- 3D Visualization Panel with a toolbar (left side): Displays the loaded molecular structure and allows interactive manipulation of the model.

- Control Panel (right side): Contains the core tools and settings, grouped into distinct sections, alongside an application exit control:
 - General: Encompasses administrative functions of the program. This includes loading files, displaying information for selected atoms, exporting results or 3D visualizations, and accessing the "About us" information.
 - Analysis: Contains the core computational tools of the software. From this section, users can perform Continuous Shape Measures (CShM), determine the molecular point group and execute symmetry analysis.
 - Style: Allows users to customize the visual rendering of the 3D model, providing access settings to modify labels, atoms representations and bond styles.
 - Exit: Control panel to safely close the application.

3.2. 3D Visualization Area:

The large panel on the left serves as the interactive molecular viewer. After loading a structure file, the molecule will be rendered in this region. The viewer supports full interactive manipulation, including:

- Rotation, translation and zooming of the molecular structure
- Single and multiple atom selection
- Visualization of geometric measurements
- Display of coordination polyhedra or symmetry elements.

The toolbar

At the top of the visualization panel, a dedicated toolbar provides quick access to essential features. You can hover your mouse over any icon to display a tooltip with its description. The available tools can be classified as:

Visualization Options

- Show/Hide hydrogens ( [h]): Toggles the visibility of hydrogen atoms in the molecular visualization. Designed to improve clarity for larger systems.
- Show/Hide Axes ( [x]): Displays or hides the Cartesian coordinates axes in the viewer. These axes help orient the molecular structure in 3D space.

Measurements

- Measure distance (d]: Measures the distances between two selected atoms.
- Measure angle (a]: Calculates the bond angle between three selected atoms.
- Measure dihedral angle (

Selection tools

- Isolate coordination environment (c]: Temporarily hides the rest of the structure to focus exclusively on the immediate coordination sphere of the selected central atom.
- Area selection (

Structure Editing and Reset

- Add bond (b]: Creates a bond between two selected atoms.
- Delete bond (Shift + *b*]: Removes a bond between two selected atoms.
- Delete atom (Delete]: Removes the currently selected atom(s) from the visualization
- Reset (Ctrl/Cmd + *z*]: Restore the molecular structure to its original state.

3.3. Control Panel

The right-side panel contains all action buttons and customization tools, organized into three main categories: General, Analysis, and Style.

General

Load File: Loads a molecular structure file into the viewer. Supported coordinate formats include currently cartesian files, (file.xyz) crystallographic information file (file.CIF) and Protein Data Bank (file.pdb). After loading, atoms and bonds are automatically rendered.

Atom Information: Displays detailed information about selected atoms, including the atom label, element type, electronegativity, coordination number and other relevant chemical properties.

Export: Allows users to export calculated data and analysis results to external files in a CSV format. Additionally, users can export high-quality images of the current molecular visualization in TIFF, PNG or JPEG formats, offering customizable resolution and background options.

About us: Displays information about the software, including developers and relevant references.

Analysis

Calculate Shape: Computes the molecular shape descriptors using the SHAPE methodology. This option compares the loaded geometry against ideal reference polyhedra and provides with the Continuous Shape Measures (CShM) for all polyhedral with the same number of vertices as defined by the user.

Get Point Group: Determines the molecular point group symmetry of the structure using symmetry-analysis libraries. The detected point group is displayed in a results window or console output.

Symmetry Measures: Calculates continuous symmetry measures (CSM) for the loaded structure. These values quantify how closely the molecule approaches ideal symmetry.

Style

Labels: Allows users to toggle and customize atom labels within the 3D viewer.

Atom: Provides setting to adjust the visual representation of atoms, such as modifying their color schemes or radii.

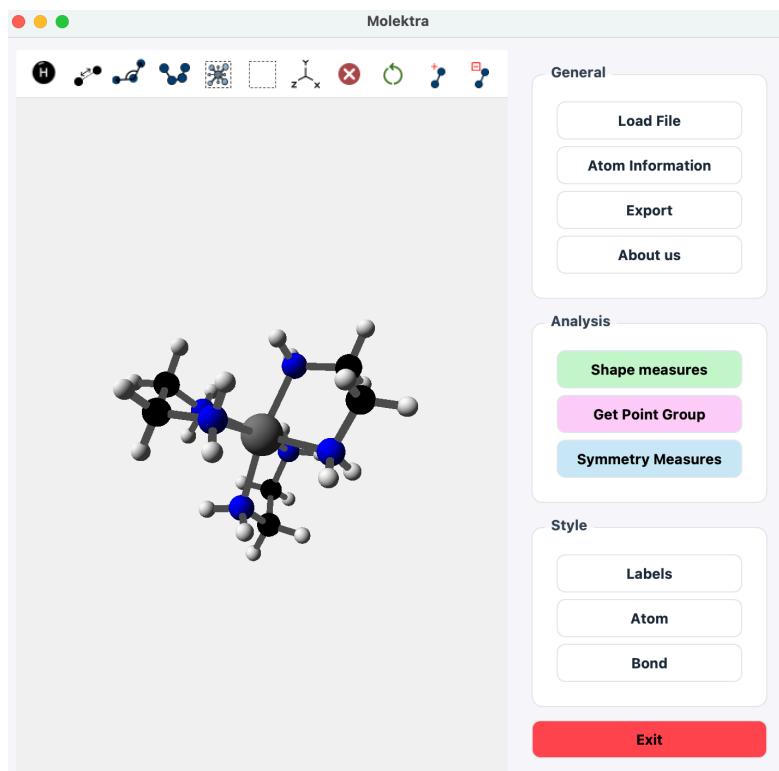
Bond: Provides settings to modify the visual style of the bond, including adjustments to cylinder thickness and color.

Application Controls

Exit: Safely closes the application.

4. Determination of the CShM values

In this section, we will work out the determination of the CShM values for the $[\text{Cu}(\text{en})_3]$ system. The typical workflow is as follows:



1. Launch the software.

Open *Molektra* to access the main interface.

2. Load the molecule.

Click the **Load File** button to open the file selection dialog. The software installation includes a folder with example files; for this case navigate to the *files_xyz* folder, select the *Cuen3.xyz* file, and click Open.

(Note: Alternatively, you can directly drag and drop the file into the 3D visualization screen).

3. Inspect the structure.

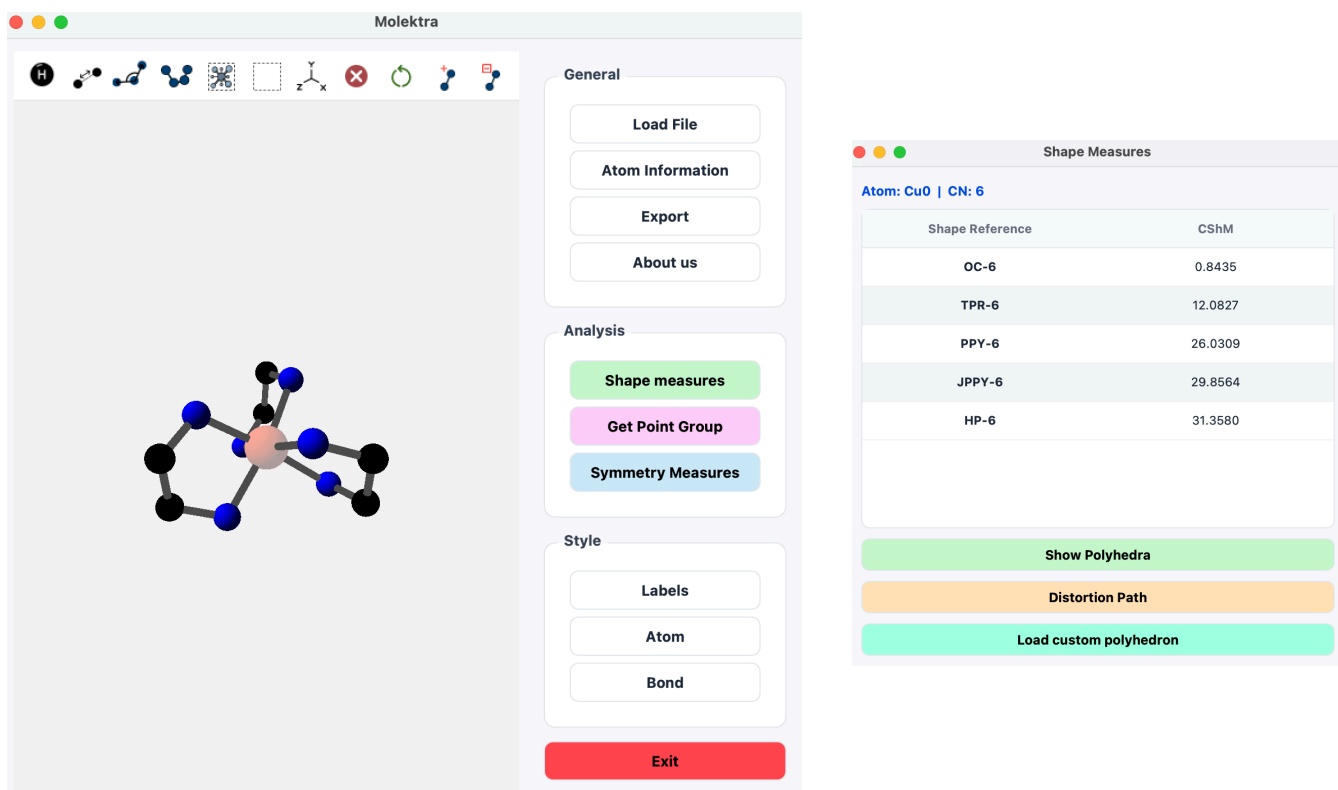
The 3D viewer will now display the loaded molecule, as show in the figure below.

4. Run the shape analysis.

Select the central Copper (Cu) atom by clicking on it in the viewer; the selected atom will be highlighted in red. Next, click the **Shape measures** button in the Analysis panel. The program automatically detects that the Cu is bound to six Nitrogens atoms, will compute the CShM for all polyhedral with 6 vertices.

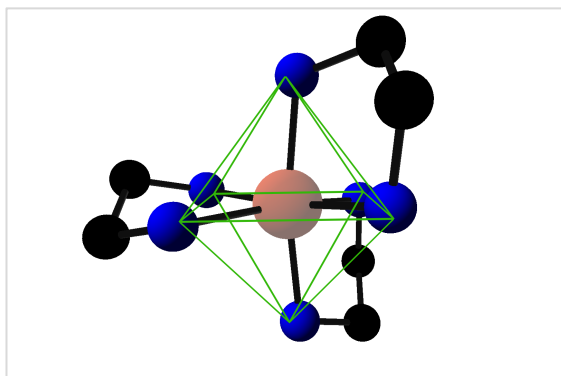
5. Interpret the results.

A pop-up “Shape Measures” window will appear, displaying a table with the CShM values for the five polyhedron with six vertices included in the SHAPE code. From the displayed values, user can clearly see that CShM(OC-6) has the lowest value, indicating that the Cu coordination environment is closest to an ideal octahedron.



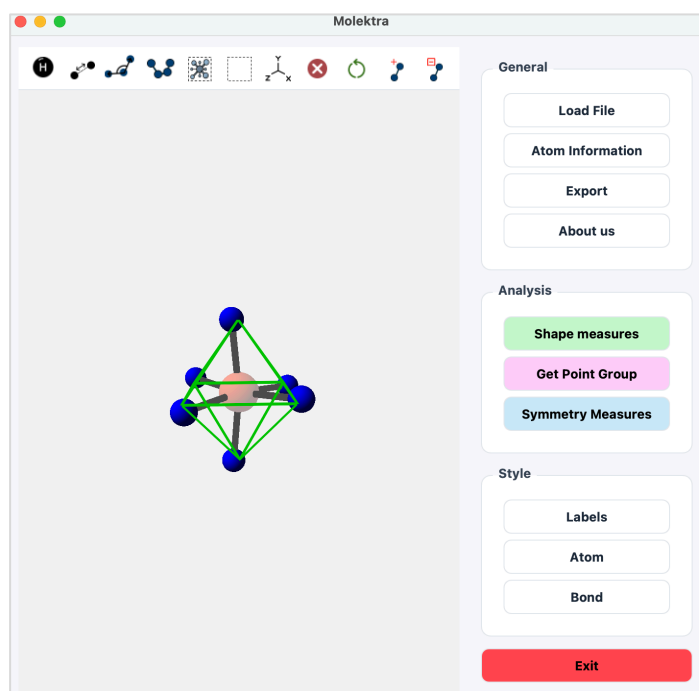
6. Visualize the ideal polyhedron

You can overlap the different ideal reference polyhedron with the molecule by using the **Show Polyhedra** button. From the different options, the user can click and select a given polyhedron and overlap it in the best possible orientation with the selected atoms. In the figure below this is illustrated for the octahedron. User can customize the color of the ideal polyhedron using the **Set polyhedron Color** button. Additionally, you can load custom reference polyhedron from an external .xyz file to compare it directly against your structure.



7. Adjust visualization options.

To improve clarity, *Molektra* provides several visualization tools in the top toolbar. For instance, you can use **Show/Hide Hydrogen** [h] to exclude hydrogen atoms from the display (as seen in previous figures). Alternatively, you can use **Isolate Coordination Environment** [c] to temporarily hide all atoms excepts the coordination sphere. This isolated view is illustrated in the figure below.



8. Export data and images.

At any point during the analysis, the user can save the results using the **Export** button. This saves the numerical measurements in a .csv file, including the labels of all evaluated polyhedral and their corresponding CShM value. If desired, user can also export a high-quality image of the molecule overlaid with the selected polyhedron.

5. Point Group and Continuous Symmetry Measures

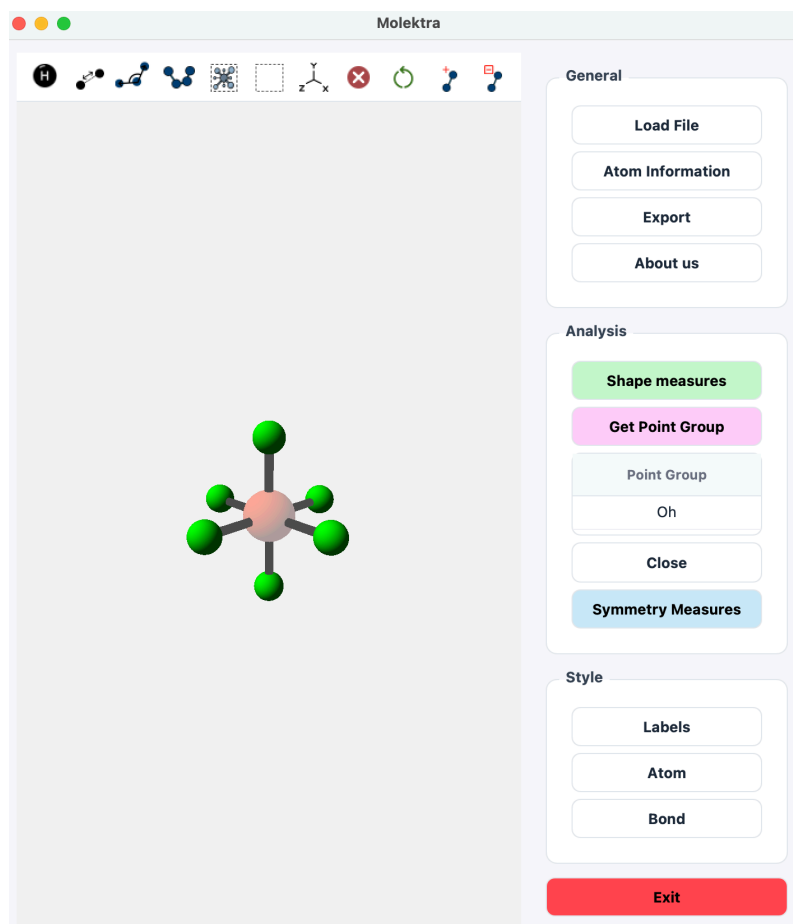
Molektra provides a comprehensive set of tools for symmetry analysis. In this example, we will determine the point group and perform a Continuous Symmetry Measures (CSM) analysis for a $[\text{MnCl}_6]^{3-}$.

1. Load the molecule.

As in the previous workflow, start by selecting and loading your molecular structure file into the 3D viewer.

2. Determine the Point Group.

From the Analysis panel in the main window, click the **Get Point Group** button. This button displays the automatically detected point molecular point group. The inset can be closed using the **Close** button. The determination of the O_h point group for the $[\text{MnCl}_6]^{3-}$ system is illustrated below.



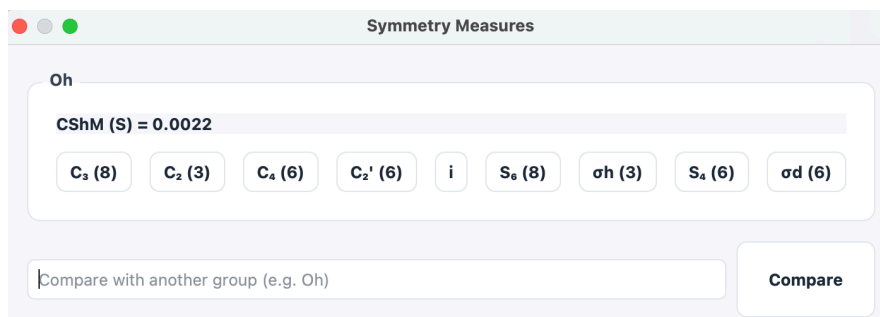
3. Run the Symmetry Measures analysis.

To perform a detailed CSM analysis, click the Symmetry Measures button. This will open a dedicated results windows containing:

- The overall CSM value for the detected Point Group.

- A detailed list of all individual symmetry operations presents within that group.
- A comparison tool at the bottom to evaluate the structure against any other point group of interest.

For the $[\text{MnCl}_6]^{3-}$ case discussed above, this menu appears as illustrated below:



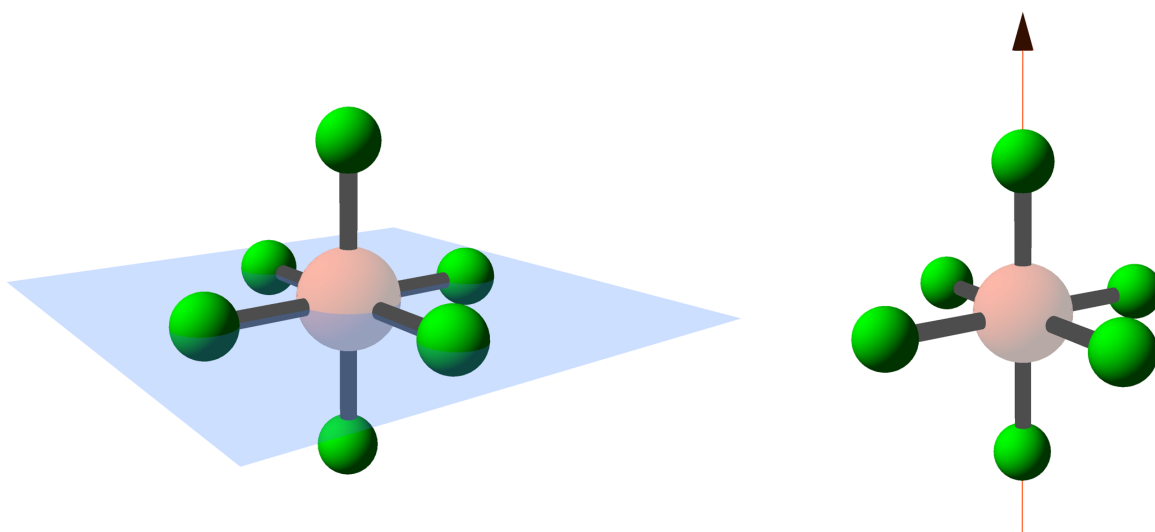
4. Inspect symmetry operations.

The results menu provides the CSM for the point group alongside all its symmetry operations. For each operation, the total number of symmetry elements is displayed in parentheses.

5. Visualize symmetry elements.

You can select any symmetry operation from the list to superimpose its corresponding symmetry element (such as rotation axes, mirror planes, or inversion center) directly onto the 3D molecule. Below are illustrated examples of how the C_4 axis the σ_h reflection plane look after being selected.

(Note: The current version of Molektra supports the visualization of only one symmetry element at a time.)



6. Keyboard Shortcuts Summary

For a faster workflow, *Molektra* supports several keyboard shortcuts for its most common tools.

Category	Action	Shortcut
Visualization	Show/hide Hydrogens	[h]
	Show/Hide Axes	[x]
	Isolate Coordination Environment	[c]
Measurements	Measure Distance	[d]
	Measure Angle	[a]
Selection & Editing	Expand Selection	[Ctrl/Cmd + E]
	Add Bond	[b]
	Delete Bond	[Shift + b]
	Delete Atom(s)/Selection	[Delete]
General	Reset Structure	[Ctrl/Cmd + z]

7. List of reference polyhedra

Vertices	Code	Label	Shape	Symmetry
2	1	L-2	Linear	D _{∞h}
	2	vT-2	Divacant tetrahedron	C _{2v}
	3	vOC-2	Tetravacant octahedron	C _{2v}
3	1	TP-3	Trigonal planar	D _{3h}
	2	vT-3	Pyramid	C _{3v}
	3	fac-vOC-3	fac-Trivacant octahedron	C _{3v}
	4	mer-vOC-3	mer-Trivacant octahedron (T-shape)	C _{2v}
4	1	SP-4	Square	D _{4h}
	2	T-4	Tetrahedron	T _d
	3	SS-4	Seesaw or sawhorse	C _{2v}
	4	vTBPY-4	Axially vacant trigonal bipyramid	C _{3v}
5	1	PP-5	Pentagon	D _{5h}
	2	vOC-5	Vacant octahedron	C _{4v}

	3	TBPY-5	Trigonal bipyramid	D _{3h}
	4	SPY-5	Square pyramid	C _{4v}
	5	JTBPY-5	Johnson trigonal bipyramid	D _{3h}
6	1	HP-6	Hexagon	D _{6h}
	2	PPY-6	Pentagonal pyramid	C _{5v}
	3	OC-6	Octahedron	O _h
	4	TPR-6	Trigonal prism	D _{3h}
	5	JPPY-5	Johnson pentagonal pyramid	C _{5v}
7	1	HP-7	Heptagon	D _{7h}
	2	HPY-7	Hexagonal pyramid	C _{6v}
	3	PBPY-7	Pentagonal bipyramid	D _{5h}
	4	COC-7	Capped octahedron	C _{3v}
	5	CTPR-7	Capped trigonal prism	C _{2v}
	6	JPBPY-7	Johnson pentagonal bipyramid	D _{5h}
	7	JETPY-7	Elongated triangular pyramid	C _{3v}
8	1	OP-8	Octagon	D _{8h}
	2	HPY-8	Heptagonal pyramid	C _{7v}
	3	HBPY-8	Hexagonal bipyramid	D _{6h}
	4	CU-8	Cube	O _h
	5	SAPR-8	Square Antiprism	D _{4d}
	6	TDD-8	Triangular dodecahedron	D _{2d}
	7	JGBF-8	Johnson - Gyrobifastigium	D _{2d}
	8	JETBPY-8	Johnson - Elongated triangular bipyramid	D _{3h}
	9	JBTP-8	Johnson - Biaugmented trigonal prism	C _{2v}
	10	BTPR-8	Biagumented trigonal prism	C _{2v}
	11	JSD-8	Snub disphenoid	D _{2d}
	12	TT-8	Triakis tetrahedron	T _d
	13	ETBPY-8	Eloganted trigonal bipyramid	D _{3h}
9	1	EP-9	Enneagon	D _{9h}
	2	OPY-9	Octagonal pyramid	C _{8v}
	3	HBPY-9	Heptagonal bipyramid	D _{7h}

	4	JTC-9	Triangular cupola	C_{3v}
	5	JCCU-9	Capped Cube (Elongated square pyramid)	C_{4v}
	6	CCU-9	Capped Cube	C_{4v}
	7	JCSAPR-9	Capped sq. Antiprism (Gyroelongated square pyramid)	C_{4v}
	8	CSAPR-9	Capped square Antiprism	C_{4v}
	9	JTCTPR-9	Tricapped trigonal prism (J51)	D_{3h}
	10	TCTPR-9	Tricapped trigonal prism	D_{3h}
	11	JTDIC-9	Tridiminished icosahedron	C_{3v}
	12	HH-9	Hula-hoop	C_{2v}
	13	MFF-9	Muffin	C_s
10	1	DP-10	Decagon	D_{10h}
	2	EPY-10	Enneagonal pyramid	C_{9v}
	3	OBPY-10	Octagonal bipyramid	D_{8h}
	4	PPR-10	Pentagonal prism	D_{5h}
	5	PAPR-10	Pentagonal antiprism	D_{5d}
	6	JBCCU-10	Bicapped cube	D_{4h}
	7	JBCSAPR-10	Bicapped square antiprism	D_{4d}
	8	JMBIC-10	Metabidiminished icosahedron	C_{2v}
	9	JATDI-10	Augmented tridiminished icosahedron	C_{3v}
	10	JSPC-10	Sphenocorona	C_{2v}
	11	SDD-10	Staggered dodecahedron	D_2
	12	TD-10	Tetradecahedron	C_{2v}
	13	HD-10	Hexadecahedron	D_{4h}
11	1	HP-11	Hendecagon	D_{11h}
	2	DPY-11	Decagonal pyramid	C_{10v}
	3	EBPY-11	Enneagonal bipyramid	D_{9h}
	4	JCPPR-11	Capped pent. Prism	C_{5v}
	5	JCAPR-11	Capped pent. Antiprism	C_{5v}
	6	JAPPR-11	Augmented pentagonal prism	C_{2v}
	7	JASPC-11	Augmented sphenocorona	C_s

12	1	DP-12	Dodecagon	D _{12h}
	2	HPY12	Hendecagonal pyramid	C _{11v}
	3	DBPY-12	Decagonal bipyramid	D _{10h}
	4	HPR-12	Hexagonal prism	D _{6h}
	5	HAPR-12	Hexagonal Antiprism	D _{6d}
	6	TT-12	Truncated tetrahedron	T _d
	7	COC-12	Cuboctahedron	O _h
	8	ACOC-12	Anticuboctahedron	D _{3h}
	9	IC-12	Icosahedron	I _h
	10	JSC-12	Square cupola	C _{4v}
	11	JEPBPY-12	Elongated pentagonal bipyramid	D _{6h}
	12	JBAPPR-12	Biaugmented pentagonal prism	C _{2v}
	13	JSPMC-12	Sphenomegacorona	C _s

8. How to cite

9. References

General references on the CShM and CSM libraries

- Cosymlib: <https://cosymlib.readthedocs.io/en/latest/>
- Posym: <https://pypi.org/project/posym/>
- Shape software: <https://www.ee.ub.edu/downloads/>

Reviews:

1. Santiago Alvarez, Pere Alemany, David Casanova, Jordi Cirera, Miquel Llunell, David Avnir “*Shape maps and polyhedral interconversion paths in transition metal chemistry*”, Coordination Chemistry Reviews, **249**(17–18), 2005, 1693-1708, <https://doi.org/10.1016/j.ccr.2005.03.031>

Methodological tools:

2. David Casanova, Jordi Cirera, Miquel Llunell, Pere Alemany, David Avnir, and Santiago Alvarez “*Minimal Distortion Pathways in Polyhedral Rearrangements*”, Journal of the American Chemical Society, **126** (6), 2004, 1755-1763
<https://doi.org/10.1021/ja036479n>

3. Jordi Cirera, Eliseo Ruiz, Santiago Alvarez “*Shape and Spin State in Four-Coordinate Transition-Metal Complexes: The Case of the d^6 Configuration*” Chemistry–A European Journal (2006) **12**(11), 3162-3167
<https://doi.org/10.1002/chem.200501516>

Reference Polyhedra:

4. Jordi Cirera, Pere Alemany, Santiago Alvarez “*Mapping the stereochemistry and symmetry of tetracoordinate transition-metal complexes*”, Chemistry–A European Journal (2004) **10**(1), 190-207
<https://doi.org/10.1002/chem.200305074>

5. Santiago Alvarez, Miquel Llunell “*Continuous symmetry measures of penta-coordinate molecules: Berry and non-Berry distortions of the trigonal bipyramid*” Journal of the Chemical Society, Dalton Transactions (2000) 29(19), 3288–3303.
<https://doi.org/10.1039/B004878J>

6. Santiago Alvarez, David Avnir, Miquel Llunell, Mark Pinsky “*Continuous symmetry maps and shape classification. The case of six-coordinated metal compounds*” New Journal of Chemistry (2002) **26** (8), 996–1009
<https://doi.org/10.1039/b200641n>

7. David Casanova, Pere Alemany, Josep M. Bofill, Santiago Alvarez “*Shape and Symmetry of Heptacoordinate Transition-Metal Complexes: Structural Trends*” Chemistry–A European Journal, (2003) **9**(6), 1281-1295
<https://doi.org/10.1002/chem.200390145>

8. David Casanova, Miquel Llunell, Pere Alemany, Santiago Alvarez “*The Rich Stereochemistry of Eight-Vertex Polyhedra: A Continuous Shape Measures Study*” Chemistry–A European Journal, (2005) **11(5)**, 1479-1494
<https://doi.org/10.1002/chem.200400799>
9. Antonio Ruiz-Martínez, David Casanova, Santiago Alvarez “*Polyhedral structures with an odd number of vertices: nine-atom clusters and supramolecular architectures*” Dalton Transactions (2008) **37(19)**, 2583–2591
<https://doi.org/10.1039/b718821h>
10. Antonio Ruiz-Martínez, Santiago Alvarez “*Stereochemistry of Compounds with Coordination Number Ten*” Chemistry–A European Journal, (2009) **15(30)**, 7470-7480
<https://doi.org/10.1002/chem.200900547>
11. Jorge Echeverría, David Casanova, Miquel Llunell, Pere Alemany, Santiago Alvarez “*Molecules and crystals with both icosahedral and cubic symmetry*” Chemical Communications (2008) **44(24)**, 2717–2725
<https://doi.org/10.1039/b719615f>